

Lab 3.2 Component:

Sets

i) Adding with sets

Input and Output:

```
a={1,2,4,53}
b={24,56,36,6,1}
a.add(7)
print(a)
```

{1, 2, 4, 7, 53}

ii) Copy with sets

Input and Output

```
c=a.copy()
print(c)
```

{1, 2, 4, 53, 7}

iii) Union, intersection, difference, symmetric difference

```
: a.union(b)
```

```
: {1, 2, 4, 6, 7, 24, 36, 53, 56}
```

```
: a.intersection(b)
```

```
: {1}
```

```
: a.difference(b) #difference means the elements of a not in b
```

```
: {2, 4, 7, 53}
```

```
: b.difference(a) #similarly elements of b not in a
```

```
: {6, 24, 36, 56}
```

```
: a.symmetric_difference(b)
```

```
: {2, 4, 6, 7, 24, 36, 53, 56}
```

iv) Discard

```
a={1,2,3,4,5,6,7,8}
a.discard(6)
print(a)

{1, 2, 3, 4, 5, 7, 8}
```

v) Update and difference update

```
a.update([9,10])
print(a)

{1, 2, 3, 4, 5, 7, 8, 9, 10}
```

```
a={23, 45, 67, 34, 56, 34, 67}
b={24,56,56,6,1}
print(a)
a.difference_update(b) #updates the set and then the result goes to a set. the difference set becomes the a set
print(a)

{34, 67, 23, 56, 45}
{34, 67, 23, 45}
```

vi) Disjoint

```
#disjoint checks for no duplicate elements in the set
a.isdisjoint(b)
```

True

vii) Disjoint and subset checking

```
a={1,2,3}
b={1,2,3,4,5,6}
print(a.issubset(b))
print(a.isdisjoint(b)) #if there is duplicate then false, if not then true
```

True

False

vii) Combined operation program

Input:

```
def list_operations():
    user_input = input("Enter elements of the list separated by spaces: ")
    my_list = user_input.split()

    while True:
        print("\n--- List Operations ---")
        print("1. Insert an element")
        print("2. Delete an element")
        print("3. Find an element")
        print("4. Display list")
        print("5. Exit")

        choice = int(input("Enter the choice: "))

        if choice == 1:
            element = input("Enter the element to insert: ")
            my_list.append(element)
            print(f'{element}' added to the list.")

        elif choice == 2:
            element = input("Enter the element to delete: ")
            if element in my_list:
                my_list.remove(element)
                print(f'{element}' deleted from the list.")
            else:
                print(f'{element}' not found in the list.")

        elif choice == 3:
            element = input("Enter the element to find: ")
            if element in my_list:
                idx = my_list.index(element)
                print(f'{element}' found at index {idx}.")
            else:
                print(f'{element}' not found in the list.")

        elif choice == 4:
            print("Current List:", my_list)

        elif choice == 5:
            print("Exiting")
            break

        else:
            print("Invalid")

list_operations()
```

Output:

Enter elements of the list separated by spaces: 4 5 6 7 8

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 1

Enter the element to insert: 9

'9' added to the list.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 2

Enter the element to delete: 8

'8' deleted from the list.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 3

Enter the element to find: 4

'4' found at index 0.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 4

Current List: ['4', '5', '6', '7', '9']

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 5

Exiting

viii) with match

```
def list_operations():
    user_input = input("Enter elements of the list separated by spaces: ")
    my_list = user_input.split()

    while True:
        print("""\n--- List Operations ---
        1. Insert an element
        2. Delete an element
        3. Find an element
        4. Display list
        5. Exit""")

        choice = input("Enter the choice: ")

        match choice:
            case "1":
                element = input("Enter the element to insert: ")
                my_list.append(element)
                print(f'{element}' added to the list.")

            case "2":
                element = input("Enter the element to delete: ")
                if element in my_list:
                    my_list.remove(element)
                    print(f'{element}' deleted from the list.")
                else:
                    print(f'{element}' not found in the list.")

            case "3":
                element = input("Enter the element to find: ")
                if element in my_list:
                    idx = my_list.index(element)
                    print(f'{element}' found at index {idx}.")
                else:
                    print(f'{element}' not found in the list.")

            case "4":
                print("Current List:", my_list)

            case "5":
                print("Exiting")
                break

            case _:
                print("Invalid")

list_operations()
```

Output:

Enter elements of the list separated by spaces: 3 4 5 6 7 8

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 1

Enter the element to insert: 9

'9' added to the list.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 2

Enter the element to delete: 8

'8' deleted from the list.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 3

Enter the element to find: 5

'5' found at index 2.

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 4

Current List: ['3', '4', '5', '6', '7', '9']

--- List Operations ---

1. Insert an element
2. Delete an element
3. Find an element
4. Display list
5. Exit

Enter the choice: 5

Exiting